

Episode 02: The Evolution of AI

Where the journey of AI began and how we have evolved so far — a high-level overview of where the AI industry stands today.

What you'll learn

- What Artificial Intelligence means
- Why AI was difficult for decades
- Why Machine Learning became necessary
- Why Deep Learning changed everything
- How Transformers changed AI
- How LLMs became popular
- Why Agentic AI is the next evolution

What is Artificial Intelligence?

AI is the science of making machines perform tasks that normally require human intelligence.



A person is considered intelligent if they can think and make decisions — if a machine can do that, it's AI.

Is it AI? — Examples

- Playing chess
- Detecting spam in email
- Recommending movies
- Writing poems
- Generating fictional images
- Driving a car (self-driving)

Why AI matters right now

- 9 out of the top 10 companies by market cap are betting heavily on AI.
- New models and tools are released every day — coding models, image generation, video, NLP.
- Every company and venture capitalist is investing in AI; a lot of research is ongoing.
- We are in the middle of a tech revolution — the biggest money in the world is flowing into AI.

Why learn the history?

The AI tools we use today have a very interesting background (~70 years of evolution). Understanding where AI came from helps you build a mental model of how it evolved.

The Early Days: 1950–1955



1950 — Can Machines Think?

Before 1950, machines were only used for computation. Alan Turing asked: “Can machines think?”

The Turing Test (Imitation Game)

A human judge asks the same questions to a human and a machine. If the judge cannot distinguish the machine from the human → the machine passes.

1955 — The Term “Artificial Intelligence” Is Coined

John McCarthy coined the term Artificial Intelligence.

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Why AI was difficult for decades

The Hype Cycle: Rise, Fall, Repeat



The 1970s–1980s: AI Winter

- Early AI researchers promised too much, but computers were too slow, there wasn't enough data, and algorithms hit a wall.
- When promises weren't kept, investors pulled out. Research slowed. Funding evaporated.
- The "AI Winter" became a period with no major breakthroughs and little motivation.

1986 — The Great Naming Debate: "Artificial" vs. "Synthetic"

Haugland asked: "Why do we use the word 'Artificial' Intelligence? Doesn't 'artificial' mean fake?"

Artificial Intelligence (AI)

- Goal: Replicate what humans do
- Approach: Copy human behavior
- How it decides: Based on training & programming
- Consciousness? No — just mimicry

Synthetic Intelligence (SI)

- Goal: Create genuinely intelligent machines
- Approach: Let machines learn independently
- How it decides: Autonomous & self-directed reasoning
- Consciousness? Possibly

The winner: "Artificial Intelligence" stuck around. Haugland's question still matters today.

The Comeback: 1997 — Deep Blue vs. Kasparov

IBM's chess-playing computer Deep Blue defeated Garry Kasparov in a match — the first time a machine beat a human in direct competition.

The Buzz (and the Reality Check)

- Analyzed permutations and combinations of chessboard positions.
- Used pure mathematics to calculate the best move.
- Played by brute force — not by understanding chess like Kasparov.
- From the outside, it looked intelligent; inside, it was running mathematics.

If something behaves intelligently, does it matter if it's not actually thinking?



Why This Mattered for AI

- Funding flooded back
- Researchers returned
- The AI winter was ending
- The narrative changed — "AI is the future"

Standing on the shoulders of giants: Alan Turing and John McCarthy built the foundation for what followed.

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Why Machine Learning became necessary

Rule-Based AI (1950s–1980s)

For the first 30 years of AI, there was only one way to make a machine “intelligent”: write rules.

Example: spam detection

if email contains "free" → spam
if email contains "\$" → spam
spam if email contains "lottery" → spam

The fatal flaw

Rules can never capture every edge case: "FR€€", "F-R-E-E", "free!!!". How many variations do you write rules for?

Machine Learning Revolution

What if we stopped writing rules? What if we showed the machine examples, and let it figure out the rules by itself?

How Machine Learning Works

1 million labeled emails → machine learns patterns → new email → “Based on what I’ve learned, this looks 85% like spam”

Trade-off: humans still had to tell the machine which features to look for.

Why Deep Learning changed everything

Automatic Feature Discovery

Machine Learning: “Look for eyes, nose, and mouth.” Deep Learning: “Here are 1 million face images” → the machine learns the features itself.



Real-World Impact

- Image Recognition — Google Photos can search “beach”.
- Face Unlock — your phone recognizes your face.
- Speech Recognition — Alexa, Siri, and Google Assistant understand you.
- Translation — Google Translate improved dramatically.

The Perfect Storm

GPU Revolution

Deep learning requires massive computation.

Internet & Data

Millions and millions of examples became available.

Corporate Investment

Companies invested billions.

What’s next?

Transformers — coming soon • LLMs — coming soon • Agentic AI — coming soon